

LESSON-16

STATISTICS

Snapshot ...

↓ Grouped Data...
↘ Ungrouped Data...

- Write the given data again in increasing order.

$$SD = \sqrt{\text{Variance}} = \sigma$$

$$\text{Variance}(\sigma^2): \frac{\sum (x_i - \bar{x})^2}{N} = \frac{\sum x_i^2}{N} - (\bar{x})^2$$

$$\text{Mean}(\bar{x}) = \frac{x_1 + x_2 + \dots + x_n}{n}$$

→ Median = Middle Value, Average of Middle two terms.

→ Mode: Repeated obs.

CF = Sum of Preceding frequencies.

$$\text{Mean Deviation About Mean} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$$

$$\text{Mean Deviation About Median} = \frac{\sum f_i |x_i - M|}{\sum f_i}$$

P.Y.Q's:

- i. Correction of Mean, Mode, Median ...

Symmetric Distribution: Mean = Mode = Median

Skewed

$$\text{Mode} = 3 \text{Median} - 2 \text{Mean}$$

$$\text{Mean: } \bar{x} = A + \left(\frac{\sum f_i d_i}{\sum f_i} \right) \times h$$

$$\text{Median: } L + \left(\frac{\frac{N}{2} - CF}{F} \right) \times h$$

• Mode:

$$L + \left(\frac{f_0 - f_1}{2f_0 - f_1 - f_2} \right) \times h$$

$$\bar{x} = \frac{\bar{x}_1 n_1 + \bar{x}_2 n_2}{n_1 + n_2}$$

$$d_1 = \bar{x} - \bar{x}_1 \quad d_2 = \bar{x} - \bar{x}_2$$

$$\sigma^2 = \frac{n_1(a_1^2 + d_1^2) + n_2(a_2^2 + d_2^2)}{n_1 + n_2}$$

$$\text{Range} = \frac{A-B}{A+B}$$

$$d_i = \frac{x_i - A}{h}$$